



# Ontology-Informed Design of Legal Visualisations

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**Abstract.** Legal regulations governing digital infrastructures and data spaces, such as the European Health Data Space (EHDS), are becoming increasingly complex and difficult to interpret, particularly for non-experts in law, including citizens and small and medium-sized enterprises. To address this challenge, we propose a method for visualising legal regulations that integrates legal ontology with legal design techniques. The method proceeds through three main stages: an ontology-informed reading of regulatory text supported by large language models (LLMs); the pre-processing of ontological constructs into structured tables enriched with explanations and examples; and the rendering of interactive concept maps using visualisation tools such as Kumu. In order to show the feasibility of the approach, we demonstrate the method using the EHDS as a case. The approach not only supports legal understanding but also contributes to the broader goal of trustworthy digital governance.

**Keywords:** Ontology · Conceptual Model · Legal Ontology · Legal Design

## 1 Introduction

Digital transformation is increasingly organised around digital infrastructures and the data spaces that run on top of them. A digital infrastructure comprises the technical artifacts (networks, protocols, platforms) and governance mechanisms that allow heterogeneous systems to interoperate at scale. However, digital infrastructures alone mainly provide connectivity and basic services. An additional organisational layer, the data space, is required to turn this connectivity into structured and trustworthy data exchange. A data space offers a governed environment where multiple parties pool and reuse data under common semantic, technical and legal rules, [12]. Its purpose is to enable cross-organisational data use while protecting fundamental rights and commercial interests.

While such shared infrastructures can emerge more or less spontaneously, they often require legislators to intervene to create them through regulations. For example, the EU's Revised Payment Services Directive (PSD2) forces banks to share account data through secure APIs, thereby giving fintechs a standardised entry point to the European payments infrastructure. The newly adopted

European Health Data Space (EHDS) Regulation establishes common rules, bodies, and technical services so that health data can be shared safely for clinical care and research.

These regulations inevitably become highly complex, as they weave together technical standards and sector-specific duties such as data protection and consumer rights, resulting in layered rule-sets that evolve over time. As infrastructures expand, so too does the complexity of the rules that govern them. For example, the PSD2 regulation has additional rules from delegated regulations and country-specific adaptations. Likewise, the EHDS works alongside the GDPR, AI Act, and MDR regulations. Navigating these multi-level, multi-source regulations can overwhelm even specialised lawyers, let alone the citizens, SMEs and public agencies expected to comply.

This growing cognitive load calls for intelligible representations of the regulatory structure. To achieve this, legal ontologies and legal design techniques can provide tools for turning dense legal text into easily digestible representations. Legal ontologies rigorously capture the legal elements and relationships of data spaces in terms of legal entities, legal positions, and legal relators. These legal ontology representations can then be translated into human-centered visualisations, narratives, and interaction patterns using legal design methods and techniques. Together, these approaches promise to reduce the interpretive burden on stakeholders, thereby assisting them in complying with the regulations.

To address the need for clarity and accessibility of complex regulations, we develop an approach based on combining legal ontology with legal design. More specifically, we propose an ontology-informed method for visualising legal regulations through concept maps, tailored to the needs of non-experts—including SMEs, patients, healthcare staff, and citizens—rather than legal professionals. To implement this method in practice, we rely on large language models (LLMs) to analyse legal texts, complemented by tools for visualising network structures.

The research presented in this paper was conducted as part of a European project focusing on supporting innovation through health data spaces. One objective of the project is the development of a “legal toolbox”, a structured set of resources designed to assist SMEs, patients, and healthcare professionals in navigating the complexities of the EHDS and related legal frameworks. Consequently, our work explicitly targets this audience of legal non-experts.

The paper is structured as follows. Section 2 provides background on the EHDS, legal design principles, analytical jurisprudence, and core legal ontologies. Section 3 elaborates the UFO-L Legal Core Ontology and its relevance to digital infrastructures. Section 4 introduces the proposed ontology-informed method for visualising legal regulations. Finally, Sect. 5 summarises the paper, discusses the implications of the method, and outlines directions for future research.

## 2 Background and Ontological Foundations

The method proposed rests primarily on the use of legal ontology, specifically the UFO-L Legal Core Ontology, [15], which is described below and in the

next section, together with underlying concepts from analytical jurisprudence. In addition, the field of legal design is outlined. The method is illustrated by applying it to the EHDS, for which a brief overview is also provided.

## 2.1 The EHDS—The European Health Data Space

The European Health Data Space (EHDS) is an initiative of the European Union designed to improve the use and governance of health data across EU member states, [2]. The EHDS builds on existing EU legislation, such as the GDPR, but also introduces specific rules for health data. It has two main goals: to give individuals better control over their personal health data, and to create a framework that supports the use of health data for research, innovation, policymaking, and public health. By setting out transparent procedures and safeguards, the EHDS seeks to build public trust in how health data is used.

For primary use, the EHDS grants citizens the right to access and share their electronic health records (EHRs) across borders. Citizens can request corrections and add their own health data, and they also have the right to restrict access to certain parts of their data. The system will rely on EU-wide formats and infrastructures to ensure interoperability between national health systems, making cross-border healthcare safer and more efficient. For secondary use, the EHDS introduces a legal and technical framework that allows researchers, public institutions, and innovators to access health data under strict conditions. The goal is to support innovation in areas such as AI, drug development, and public health responses, while respecting patients' rights.

The regulation introduces several new legal actors, including National Contact Points for Digital Health (for primary use coordination), Health Data Access Bodies (for secondary use governance), and the European Health Data Space Board (to ensure consistent implementation across the EU). The EHDS regulation was adopted in January 2025 and entered into force in March 2025, with implementation expected to be phased in over several years.

## 2.2 Legal Design

Legal design is an interdisciplinary field that integrates methods and techniques from law, design thinking and information technology to improve how legal information is presented. Its main goal is to make the information clearer, easier to understand and more usable by focusing on the needs of people interacting with the law, [9]. For these purposes, legal design applies techniques such as plain language, visual communication, user stories, iconography, and decision trees, [19]. This makes legal materials not only more accessible but also more likely to be trusted and used correctly.

Legal design techniques are helpful for people who often find legal materials difficult to access, such as citizens trying to understand not only laws but also other legal documents, including contracts and court decisions. Small businesses can also benefit, especially when they face complex rules but lack in-house legal teams. For example, under the EHDS regulation, visual maps and user stories

can help patients understand the rules governing their access to their own data, as well as their rights to restrict data access for care providers. Analogously, suppliers of EHR systems can benefit from an accessible overview of the EHDS requirements, allowing them to address all of these in an early stage, rather than facing costly redesigns later.

### 2.3 Analytical Jurisprudence

Analytical jurisprudence, as a tradition within legal theory, has long sought to systematically clarify core legal concepts and their relationships. By analysing these notions with philosophical rigour and bracketing moral or policy considerations, it aims to supply precise conceptual frameworks that can be applied across diverse legal areas. An early work in this tradition was Wesley Newcomb Hohfeld's analysis of legal positions, [22], which distinguished four basic types: claim-rights, privileges, powers, and immunities.

- *Claim-right*. If an actor A has a claim-right against another actor B to do (or not do) some act, then B has a correlative duty toward A to fulfill that claim by doing (or not doing) the act.
- *Privilege*. If an actor A has a privilege to do an act, it means A is under no duty not to do the act. The correlative of A's privilege is that any other actor lacks a claim-right to require A to do otherwise.
- *Power*. A power is the legal ability to alter legal positions, i.e., to create, modify, or extinguish claim-rights, duties, privileges or other powers. Whenever A has a power regarding B, B correspondingly has a liability, meaning B's legal position can be changed by A's exercise of the power.
- *Immunity*. An immunity is the flipside of a power: it is an exemption from having one's legal status altered by another. If A has an immunity against B, it means B lacks the power to change A's legal position in some respect.

This analysis of basic legal positions makes it possible to disaggregate complex legal relations into combinations of legal positions. For instance, ownership of property can be analysed as a bundle of legal positions (claim-rights against others' interference, privileges to use, powers to transfer, etc.).

Herbert L.A. Hart was also a central figure in analytical jurisprudence, but with a broader focus. Rather than analysing individual legal relations in isolation, Hart aimed to explain the nature of a legal system, [20]. One of his major contributions was the concept of secondary rules, i.e., rules that empower actors to create, modify, or extinguish primary rules, i.e., rules that impose or remove duties. Examples include rules that confer legislative power, or the rules for making wills or contracts. There is a correspondence between these notions and Hohfeld's typology: privileges and claim-rights map onto primary rules (governing conduct), while powers and immunities map onto secondary rules (governing legal change). Unlike Hohfeld, who only classified legal relations already assumed to exist, Hart also explained how such relations arise from a structured system of rules. For instance, a contract or statute is valid only if created in accordance with applicable secondary rules.

Legal Institutional Theory, as developed by MacCormick and Weinberger, encompasses the key contributions of both Hohfeld and Hart. However, it differs from them by conceiving of law as an “institutional normative order”—a system of norms founded on, and shaped by, social institutions, [25]. Norms can be understood logically (in terms of their content and relations to other norms) and at the same time as observable social phenomena (in terms of being followed and enforced in institutional settings). Legal Institutional Theory diverges from Hohfeld by being less focused on the pure classification of legal rules and more on explanation: how those rules come about through institutional norms and what role they play in practical reasoning. Practical legal reasoning means that when applying the law, one must often interpret rules or precedents purposively, weigh principles, and ensure coherence in the law as a whole; solely applying legal rules is sometimes insufficient for deciding cases. Thus, judges also utilise principles and reasoned arguments concerning the purpose of legal rules. And these arguments are guided by the traditions and values of specific institutions.

## 2.4 Core Legal Ontologies

Core legal ontologies are conceptual frameworks that aim to capture the basic conceptual structure of law in a domain-independent way. Their primary purpose is to support the formal representation of legal knowledge in a consistent and uniform manner, thereby enabling interoperability among legal representations and providing a basis for designing legal domain ontologies. A well-established core legal ontology is the Legal Knowledge Interchange Format (LKIF-Core), [21], which was informed by multiple foundational ontologies and provides a comprehensive set of legal notions to support practical legal reasoning. Another example is CLO, [13], based on the DOLCE foundational ontology. Similarly to CLO, UFO-L, [16], builds on a solid foundational ontology: The Unified Foundational Ontology, UFO, [18]. UFO-L takes a relational perspective on core legal concepts based on the work by Hohfeld, [22] and Alexy, [5]. We have chosen UFO-L as the basis of our work, as it offers a relational and event-aware structure that is able to capture the multi-party, evolving legal relationships that are characteristic of digital infrastructures and data spaces.

In preparation for the description of UFO-L in the next section, we here review some key constructs of UFO. UFO makes a key distinction between individuals and universals. *Individuals* in UFO are entities that possess a unique identity, while *universals* are abstract patterns of features that can be realised in one or several individuals. Individuals can be divided into endurants and events, see Fig. 1. An *endurant* is an individual that is wholly present whenever it is present, e.g., a mountain or a car. An *event*, on the other hand, is made up of temporal parts, which are not all present whenever the event is present.

Endurants can be specialised into substantials and moments. *Substantials* are existentially-independent endurants, e.g., a tree or a person. In contrast, *moments* are individuals that inhere in other individuals, meaning that they are existentially-dependent on their bearers, e.g., the smile of a person or the marriage between two spouses. Moments can be either intrinsic moments or



relators. *Intrinsic moments* are moments that inhere in a single endurant, e.g., the colour of a car, while *relators* are dependent on two or more endurants, e.g., a marriage.

## 2.5 Visualisation

Legal conceptual models on the level of granularity of legal positions quickly become very large. The EHDS regulation is around 60,000 words long, including annexes, composed by 105 articles, most of which contain several numbered paragraphs. While a model does not have to cover the complete regulation to be useful, also a limited model will be rather large. How to visually represent the model becomes crucial if lay users are to increase their understanding.

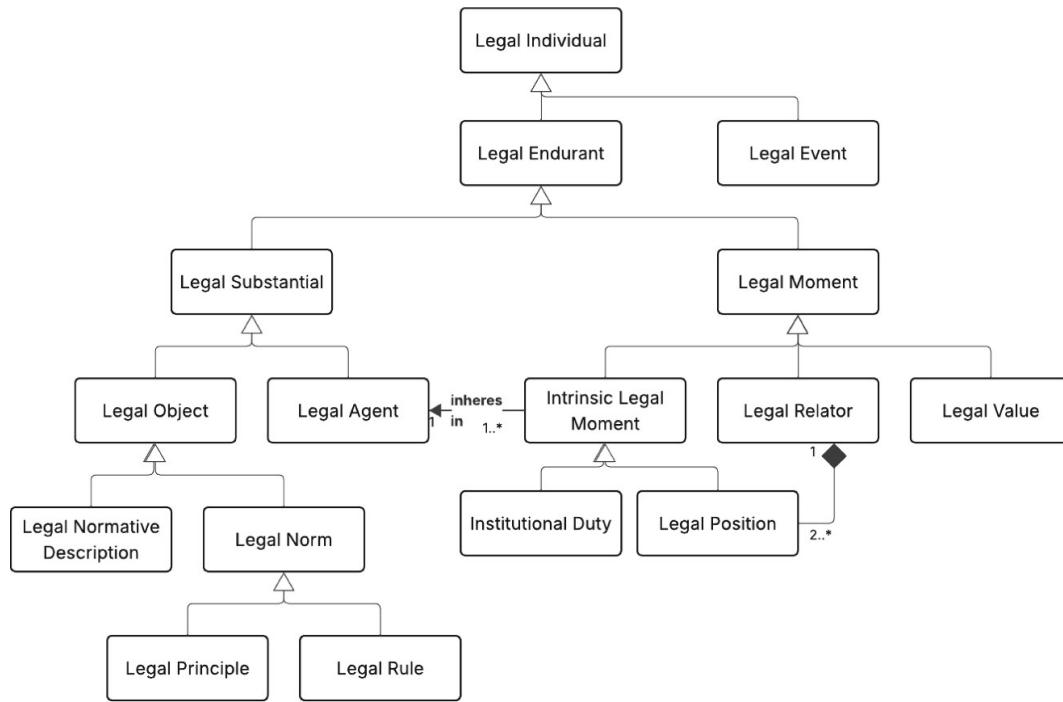
Visualisations can bring new insights to complex information, such as legal texts made more accessible. Legal visualisation has mostly focused on the relations between legal texts, for example, citations, but has also been used to understand the relations between concepts [26]. Legal texts can be analysed with natural language processing (NLP) to create visualisations.

In visualisation of large models, secondary notation becomes increasingly important. As explained by Moody, [27], while the primary notation of a modelling language is defined by its formal rules, the secondary (informal) notation provides other visual clues to how interpret a diagram. The informality of secondary notation does not make it unimportant, however. In fact, Ghiran et al. propose that secondary notation should be understood as an “explicit building block of a modelling method” [14, p. 1], including dynamic rendering through visualisation tools.

As argued by Cerioli et al. [7], for a network visualization to be effective, users must be able to navigate across multiple object types, including actors, rules, contracts, and other legal entities. To support this, the visualization should employ metaphors that resonate with users’ mental models. Cerioli et al. propose a map-based approach, which they term *wayfinding*, enabling users to move in and out of the map, discern direct and indirect connections, and explore the broader context. In the legal domain, such a map metaphor is particularly valuable: as Ruhl and Katz [29] note, it provides an intuitive framework for designing *legal maps*, allowing practitioners and scholars to navigate complex legal structures and relationships more effectively.

## 3 UFO-L and Digital Infrastructure Regulations

This section provides an overview of UFO-L, with particular attention to its relevance for digital infrastructure regulations. The overview integrates elements from the models presented in [16, 17], extended by two additional classes. A representation of UFO-L is shown in Fig. 1 as a UML class diagram.



**Fig. 1.** A Fragment of UFO-L

### 3.1 Legal Agents and Legal Objects

A *legal individual* is an individual that is recognised and regulated by a legal system. Legal individuals are divided into legal events and legal endurants, which in turn are divided into legal substantials and legal moments. A *legal agent* is a legal substantial recognised by the legal system as capable of having rights, privileges and other legal positions. They can typically own property, enter into legal relations, or be held responsible. Legal agents can be natural persons (corresponding to human beings) or juridical persons, such as companies, associations, and authorities. Some examples of legal agents in the EHDS are patients, health professionals, member states, and national contact points.

A *legal object* is a legal substantial that can be the target of primary legal positions, i.e., claim-rights and privileges. Legal objects are what these legal positions are about, but they cannot themselves hold legal positions. Instead, legal agents hold legal positions concerning legal objects. An example is that a patient (legal agent) has the right to access their health data (legal object), and a hospital (legal agent) has the duty to provide it. Other examples of legal objects from the EHDS are personal e-health data, electronic health record (EHR) systems, metadata, standardised data formats, and datasets.

### 3.2 Legal Relators and Legal Positions

A *legal moment* is a legal endurant that is dependent on at least one legal agent. A *legal relator* is a legal moment dependent on two or more legal agents, e.g., a marriage that is dependent on two legal agents. In contrast, an *intrinsic legal*

*moment* inheres in only one legal agent. In UFO-L, a *legal position* is modelled as an intrinsic legal moment that is externally dependent on a legal agent. For example, a duty for A to pay an amount of money to B is viewed as an intrinsic legal moment that inheres in A, but it is also externally dependent on B.

Legal positions are part of legal relators. For example, a marriage relator can include a duty of one spouse to financially support the other spouse (a legal position) as one of its parts. It can also be noted that Hohfeld views a legal position as a two-place relation between two agents. However, UFO-L, as well as others, [5], views a legal position as a three-place relation that also makes explicit what the relation concerns. Primary positions (rights and privileges) are about legal objects, while secondary positions (powers and immunities) are about modifying other positions.

Every legal relator (as well as other legal endurants) follows a lifecycle, beginning at a specific point in time and ending at another. In the terminology of [25], an *institutive rule* specifies the conditions under which a relator comes into existence. For example, a marriage is created when the prescribed wedding has been performed and the prerequisites (e.g., consent, witnesses, and registration) are fulfilled. Analogously, a *terminative rule* specifies the conditions under which a given relator ceases to exist. In a law text, these rules are typically not detailed procedural descriptions but instead constraints outlining necessary conditions.

### 3.3 Institutional Duties

A key feature of Hohfeld’s framework is its strict correlativity: every legal position has a correlative. For instance, if one agent has a duty, another agent must hold a claim-right. Yet, in practice, there are situations where the identity of a specific claim-right holder is unclear or indirect. As MacCormick, [24], observes, “not all duties are relevantly ‘owed to’ determinate other individuals.” Public authorities, for example, may bear duties such as safeguarding public safety or protecting the environment, where no clearly identifiable individuals hold the corresponding claim-right. Instead, these duties are generally understood as owed to “the public”, “society at large”, or even “the environment”.

Such institutional duties are ubiquitous in regulations of digital infrastructures and data spaces, as these are designed to ensure the systemic functioning of digital ecosystems. And they do this by imposing duties aimed at securing system properties, such as interoperability, stability, and trust. Thereby, these duties, such as adhering to technical standards or contributing to governance mechanisms, serve the integrity of the system rather than corresponding directly to the claim-rights of specific agents. The EHDS also illustrates how such regulations often pursue public interest goals, such as facilitating cross-border health research, which benefits society as a whole rather than specific individuals. An example of an institutional duty from Article 19 in the EHDS is “Each digital health authority shall be entrusted with the following tasks ... ensuring the implementation at national level of the European electronic health record exchange format”. In Fig. 1, we have added a class “Institutional Duty”, which is not in UFO-L.



### 3.4 Legal Norms, Legal Values, and Legal Normative Descriptions

*Legal norms* are standards that tell agents what behaviours are required, permitted, prohibited, or enabled in a legal system. A *legal rule* is a particular kind of legal norm characterised by a high degree of specificity and an all-or-nothing mode of application. For example, Article 3.2 of the EHDS states, “Natural persons . . . shall have the right to download free of charge an electronic copy of at least the personal electronic health data in the priority categories referred to in Article 14 related to those natural persons”. This rule grants people in the EU access to their health data.

Because of their clarity and specificity, legal rules provide certainty and predictability, but can be difficult to apply in hard cases. In contrast, *legal principles* are norms that guide legal interpretation and influence the application of legal rules. Dworkin distinguished principles from rules by noting that principles are not applied in an all-or-nothing fashion; instead, they have a dimension of weight or importance, [11, p. 77–78]. A principle provides a reason that may influence, but not determine, a decision, and it can coexist with competing principles. An example, often referred to in the EHDS, is the principle of data minimisation in the GDPR, which requires that personal data collected and processed must be adequate, relevant, and limited to what is necessary in relation to the purposes for which it is processed.

*Legal values* are foundational ethical and social ideals within a society that a legal system intends to uphold. They represent the “general, basic and framework norms upon which a legal system is built”, [10]. Typical examples of legal values are order, fairness, equality, and freedom. Some legal values relevant for the EHDS are patient autonomy, privacy, trust, accountability, and innovativeness.

Legal values and legal principles differ in their functions for the legal system. The primary function of principles lies within the legal system, as they guide interpretation and ensure consistency of rules—legal principles are tools for legal reasoning and application. In contrast, the primary function of legal values is justificatory, as they provide the underlying rationale for the norms of the legal system, grounding it in the broader ethical and social commitments of a society—legal values answer the ‘why’. Legal rules and legal principles are included in UFO-L but not legal values, and we have added a class to Fig. 1.

A legal norm is linguistically represented through a *legal normative description*, which is a linguistic object, such as an article or a paragraph in a legal text. The distinction between norm and description helps differentiate between normative content (the “ought”) and legal documentation (the symbolic expression of that ought).

## 4 A Method for Visualising Legal Regulations

We propose a method for visualising legal regulations using concept maps, where the map components correspond to the ontological constructs defined in UFO-L. The method includes three main activities that are described in the following subsections. We also describe the specific tools we used to implement the

method. Doing so demonstrates the feasibility of the method by showing that widely available tools can operationalise each step in the method. It also supports transferability: if future researchers choose to use alternative tools, the pairing of method steps with concrete tools clarifies the functional requirements that any replacements must meet. For an overview of the method, see Fig. 2.

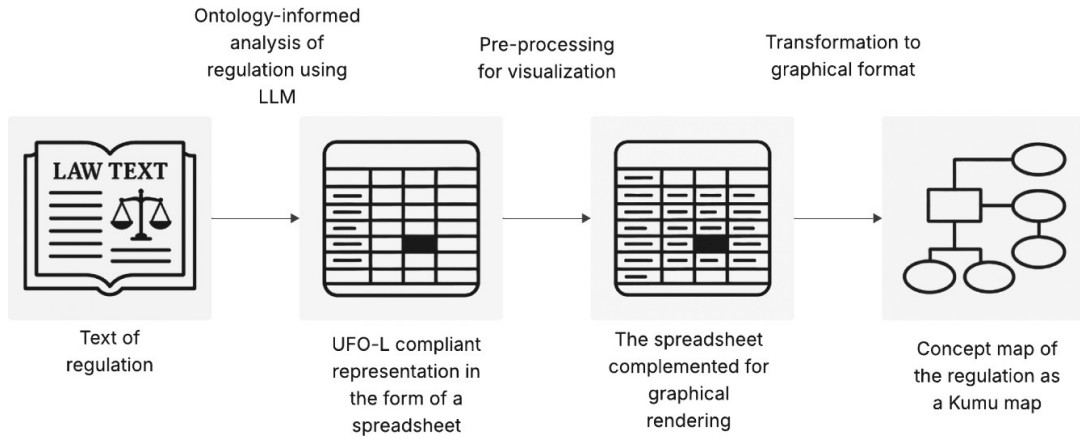
#### 4.1 Ontology-Informed Analysis of Regulations with LLM

The method begins with an ontology-informed analysis of the regulatory text. The goal of this step is to re-express the text in UFO-L terms, with a focus on identifying the relevant universals it contains. The step proceeds through a systematic reading and interpretation of the text. In this work, LLM tools can provide support through suggestions on interpretations of articles and paragraphs, as well as feedback on proposed interpretations. For this purpose, we have developed several prompts. First, a priming instruction was given explaining the terms used in the succeeding prompts. Second, for each paragraph and legal relation contained in it, the type of the legal relation was identified. Third, for each paragraph and legal relation, a specific prompt, based on the type of the relation, was given to extract its contents. Three examples of specific prompts are shown below.

- Read Paragraph X, which concerns a *claim-right*. Describe it in a table format with the following columns: Rights Holder, Duty Bearer, Original Wording, Action, Legal Object, Condition, Purpose.
- Read Paragraph X, which concerns an *institutional duty*. Describe it in a table format with the following columns: Duty Holder, Original Wording, Action, Legal Object, Condition, Purpose.
- Read Paragraph X, which concerns a *power*. Describe it in a table format with the following columns: Power Holder, Subjection Holder, Original Wording, Legal Position Created or Modified by the Power, Action, Legal Object, Condition, Purpose.

For the first prompt, Rights Holder and Duty Bearer are universals whose instances are legal agents; Original Wording is the original text of the paragraph; Action is the act that the claim-right concerns; Legal Object is a universal whose instances are legal objects; Condition describes a prerequisite or a restriction of the application of the claim-right; and Purpose is about the rationale of the claim-right.

Our experience using an LLM tool (ChatGPT 4.5 and o3) as support has been generally promising, with the generated responses being both accurate and practically useful in almost all cases. However, outputs still require manual verification and, occasionally, some rephrasing. Beyond generating responses for our pre-defined prompts, the LLM tool has been valuable as a dialogue partner in more complex analytical tasks, particularly those requiring iterative work on interpretations.



**Fig. 2.** Overview of the Method for Visualising Legal Regulations

## 4.2 Pre-processing for Visualisation

The second step involves pre-processing the results from the previous step into a tabular format—specifically, a spreadsheet—that can be used by a tool to render the concept map. Three sub-steps are included:

- A table is created for universals that contain instances of Legal Agent (Legal Object, Role, Claim-right, Privilege, Power, Immunity, Institutional Duty, Legal Relator, and Value). The columns of the tables may vary, as suggested by the above prompts.
- For each table above, two columns Explanation and Example are added. Explanation is a paraphrase of an original text in the regulation in plain language, and Example is a concrete case illustrating an item in easily understandable language. An LLM tool assists in generating these paraphrases.
- A table Connections is created for all relations between universals, in particular, for inheres-in and external-dependency relations.

These steps are informed by the following principles for visual notations by [27]:

*Semiotic Clarity.* There should be a 1:1 correspondence between ontological constructs and graphical symbols. This principle is the basis of our visualisation approach; the rows of the tables corresponding to universals will become the nodes in the concept map.

*Dual Coding.* Text annotations should be used to complement graphical symbols to offer additional information. To operationalise this principle, we have prepared clear explanations and illustrative examples using accessible language, guided by the principles of plain legal language [4].

*Graphic Economy.* The number of different symbols should be kept small so that users will not become overwhelmed and struggle to remember the meanings of the symbols. We have primarily addressed graphic economy through symbol deficit, i.e., by deliberately omitting graphical representations for certain ontological notions. In particular, we have not graphically modelled normative descriptions (articles and paragraphs), but instead included these only in text annotations.

### 4.3 Transformation to Graphical Form

The third step transforms the tables obtained in the previous step into a concept map. The sub-steps specify how the nodes are to be rendered:

- Each row in the tables for Legal Agent (e.g. “Digital Health Authority”, “Health Professional”, “Natural Person”) and Legal Object (e.g., “Personal Electronic Health Data”) gives rise to a large node with a unique icon.
- Each row in the tables for Claim-right (e.g. “Right to Have Access”), Privilege, Power (e.g. “Power to Restrict”), Immunity, and Institutional Duty, gives rise to a small node with the same icon.
- Each row in the table for Legal Relator (e.g. “Restriction of Access”) gives rise to a mid-sized node with the same icon.
- Each row in the table for Value (e.g. “Trust”) gives rise to a large node with the same icon.
- Each row in the Connections table gives rise to an edge between the related nodes.

For the actual rendering of the map, we have used Kumu, [3], which is a web-based tool designed to support the analysis and visualisation of complex systems through interactive network maps. It enables users to model and explore relationships among diverse elements such as stakeholders, concepts, and policies. Kumu allows for customisation: users can employ visual variables like colour, size, shape, icons, and layout to highlight key elements and distinguish between different types of nodes and edges. The tool also supports filtering and zooming functionality, enabling users to explore a map interactively.

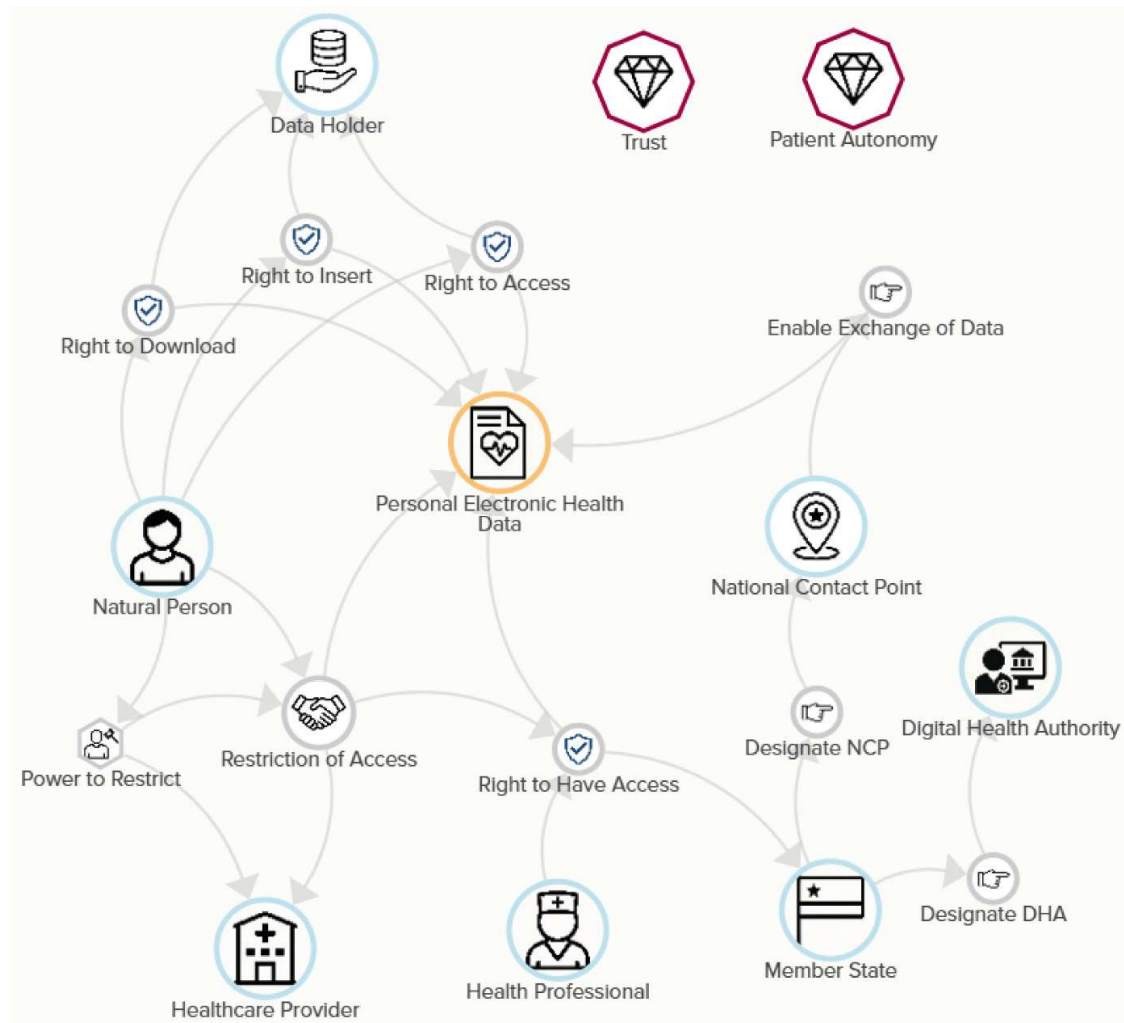
In this step, we also draw on the visualisation principles in [27]:

*Semantic Transparency.* The appearance of a symbol should suggest its meaning, preferably so that it can be directly perceived. To this end, we have selected icons that are both intuitive and familiar to the intended audience. The icons have been informed by DaPIS: the Data Protection Icon Set, [1].

*Perceptual Discriminability.* Different symbols should be clearly distinguishable from each other, so that users can easily differentiate between various elements on the map. We have addressed this issue by employing different shapes and colours, as well as drawing icons in a minimalistic style with high internal contrast.

*Visual Expressiveness.* The full range and capacity of visual variables (such as shape, colour, size, and texture) should be utilised to support both semantic transparency and perceptual discriminability. We have addressed this principle using Kumu’s features to render map elements. For example, we use larger sizes and distinct icons for legal agents to highlight their importance and ensure they are clearly distinguishable from other map elements.

*Manageable Complexity.* The visualisation should include mechanisms for handling complexity, such as modularisation and hierarchical structuring, to ensure that users are not overwhelmed by large and complex maps. For this purpose, we rely on Kumu’s functionality, which offers zooming and filtering to focus on specific elements as well as grouping to divide large maps into segments.



**Fig. 3.** A Fragment of a Concept Map for EHDS Chapter 2

For example, certain edges in a map can be hidden by default in order to reduce clutter, as shown in Fig. 3, where the edges from the Value nodes are initially suppressed but can be made visible interactively if so desired.

Figure 3 shows a small fragment of a concept map based on Chap. 2 of the EHDS, which addresses the primary use of health data. (Large circles with a blue border represent legal agents; large circles with an orange border represent legal objects; octagons represent values; medium-sized circles with a handshake icon represent legal relators; and all other nodes represent legal positions.) The figure illustrates the output from a method application. As a static image, it cannot display the map’s interactive features, but a live, navigable version is available online<sup>1</sup>. Figure 4 displays a pop-up window that contains a text annotation for a node in the map.

<sup>1</sup> <https://embed.kumu.io/cf43985dcc788d7b433ec3bd22d0f8fb>.



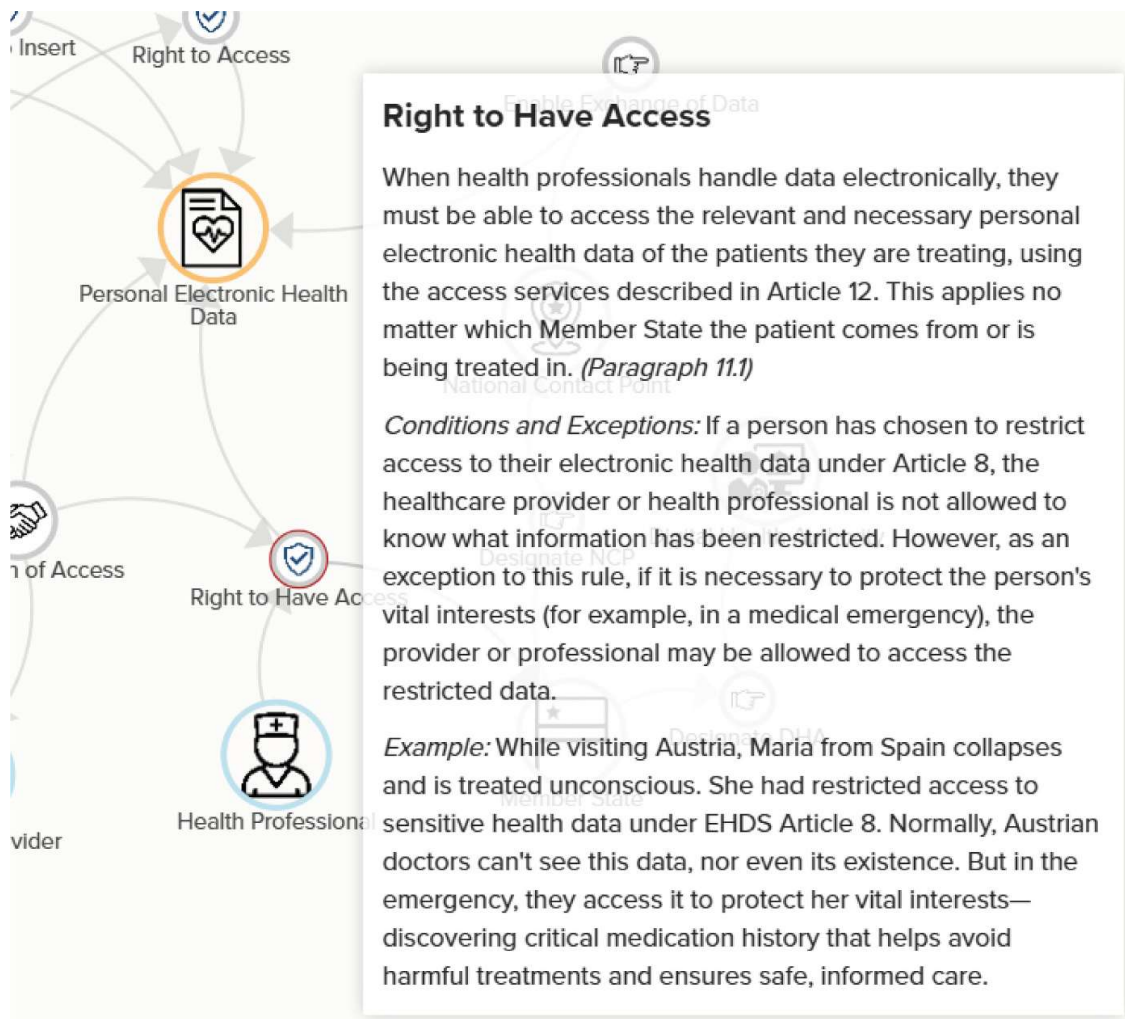


Fig. 4. A popup window containing a textual annotation

## 5 Discussion

We have proposed an ontology-informed method for visualising complex legal regulations using concept maps, demonstrated through a case study of the EHDS regulation. By applying UFO-L, we showed how to systematically analyse legal texts to identify and represent legal core elements. To support and streamline this analysis, we used LLMs, which helped interpret statutory texts and generate easily understandable paraphrases and illustrative examples. The resulting concept maps facilitate navigation and interpretation of regulatory material, particularly benefiting legal non-experts. While the proposed method has been developed with a primary focus on digital infrastructures and data spaces, it has broader applicability across other legal domains.

Methodologically, our research process can be seen as an instance of design science research, [28]. Using ontology as a knowledge base, we have addressed a persistent and widely recognised problem: making legal material comprehensible for non-experts. We have developed a method for creating concept maps of complex regulations, which we have demonstrated in a case study, applying it



to a part of Chapter 2 of the EHDS. A thorough evaluation of the method is a topic for future research.

Scholarship on the visualisation of legal materials has produced a comprehensive body of work, much of it centered on network graphs that depict how statutes or regulations cross-reference and influence one another. For example, Koniaris et al. introduced the Legislation Network and applied it to a corpus of EU law, in which each legislative act appears as a node and citations form the connecting edges [23]. A distinct line of inquiry examines the logical form of norms: Tosatto et al. proposed a graphical representation that encodes premises, deontic operators and conditional clauses, allowing researchers to trace the inferential architecture of articles and paragraphs, [8].

Although these approaches support detailed inferential analysis valuable to legal professionals, their granularity may hinder lay comprehension. Closer to our objectives are studies that visualise networks of agents and their legal positions without delving into the syntactic minutiae of the underlying legal text. The pioneering efforts of Burkhardt and Nazemi, [6], as well as Griffo et al., [15], illustrate this approach; however, their models still presuppose that users are familiar with legal constructs, targeting legal professionals rather than non-expert stakeholders.

Our method for converting natural language legal text to structured legal relations can be considered a form of ontology-guided prompting. A limitation of our proposed approach is that we did not engineer the prompts according to any systematic methods or frameworks. This likely affects the quality of the LLM-generated legal relations. In future studies, a possible approach could be to employ schema-guided prompting, in other words, that both the input is given in a structured form, for example, JavaScript Object Notation (JSON). This would lead to improved predictability.

For future work, we plan to consolidate and extend the use of UFO-L. While UFO-L has proven to be a robust foundation for analysing legal texts and developing concept maps, we have not fully utilised its potential. Specifically, we have not accounted for legal events, and more importantly, we have focused solely on legal universals in our analysis without addressing legal individuals. One of the challenges with modelling legal systems is that they offer an intriguing interaction between universals and individuals. For example, an individual law (EHDS) creating a new universal (National Contact Point), which in turn will be instantiated through a combination of national regulations and tangible information infrastructure design. Another universal created by the EHDS—Personal Electronic Health Data—will have a massive number of instances from the moment it is created, since much data will fit the definition in the EHDS.

Moreover, the concept of legal value merits deeper examination, for which Sales’s work offers a promising foundation, [30]. As indicated in Fig. 3, the relationships between Values and other elements remain challenging. One reason is that Values interact with the larger system rather than its individual components. One approach, then, would be to represent the larger system and connect

it with its parts. However, a weakness of this solution is that legal systems have fuzzy borders, making it rather arbitrary where to draw the line.

Another topic for further research is to apply ontology-informed analysis to legal design techniques other than concept maps. Process diagrams and scenario narratives, for instance, could give non-expert stakeholders intuitive guidance, and they can be generated methodically from the very analysis used here. Thus, one ontological analysis could underpin several complementary forms of representation.

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